EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 7B

1. Steady and Unsteady States

Consider a system whose state evolves according to $\mathbf{x}[k+1] = \mathbf{M}\mathbf{x}[k]$, where

$$\mathbf{M} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & -2 \\ 0 & 0 & 2 \end{bmatrix}.$$

In this problem, we will use the eigenvalues and eigenvectors of $\mathbf{M}$ to predict the long-term behavior of the system.

(a) For each eigenvalue below, find a nonzero eigenvector $\mathbf{v}_i$ of $\mathbf{M}$ and give its eigenspace $\text{span}(\mathbf{v}_i)$:

i. $\lambda_1 = 1$
ii. $\lambda_2 = 2$
iii. $\lambda_3 = \frac{1}{2}$

Answer: Since $\mathbf{M}$ is upper triangular, so is $\mathbf{M} - \lambda\mathbf{I}$, and we can solve $(\mathbf{M} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ for $\mathbf{v} = \begin{bmatrix} x & y & z \end{bmatrix}^\top$ by back substitution, starting from the last row.

i. $\lambda = 1$:

$$(\mathbf{M} - \mathbf{I})\mathbf{v} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & -2 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{v} = \mathbf{0}$$

The last row gives $z = 0$. The first row then gives $-\frac{1}{2}x + \frac{1}{2}y = 0$, so $x = y$, and y is free.

$$\mathbf{v}_1 = \alpha \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \alpha \in \mathbb{R}$$

ii. $\lambda = 2$:

$$(\mathbf{M} - 2\mathbf{I})\mathbf{v} = \begin{bmatrix} -\frac{3}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \mathbf{v} = \mathbf{0}$$

The last row is all zeros, so z is free. The second row gives $y = -2z$. The first row gives $-\frac{3}{2}x + \frac{1}{2}(-2z) - \frac{1}{2}z = 0$, so $x = -z$.

$$\mathbf{v}_2 = \beta \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}, \beta \in \mathbb{R}$$

iii. $\lambda = \frac{1}{2}$:

$$(\mathbf{M} - \tfrac{1}{2}\mathbf{I})\mathbf{v} = \begin{bmatrix} 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & -2 \\ 0 & 0 & \frac{3}{2} \end{bmatrix} \mathbf{v} = \mathbf{0}$$

UCB EE 66, Fall 2026, Discussion 7B, All Rights Reserved. This may not be publicly shared without explicit permission.

The last row gives $z = 0$, and the second row then gives $y = 0$. The first column is all zeros, so x is free.

$$\mathbf{v}_3 = \gamma \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \gamma \in \mathbb{R}$$

Meta/Answer: (8 min)

(b) Define $\mathbf{x} = \alpha\mathbf{v}_1 + \beta\mathbf{v}_2 + \gamma\mathbf{v}_3$, a linear combination of the eigenvectors. For each of the cases in the table, determine if

$$\lim_{n\to\infty} \mathbf{M}^n\mathbf{x}$$

converges. If it does, what does it converge to?

α	β	γ	Converges?	$\lim_{n\to\infty} \mathbf{M}^n\mathbf{x}$
0	0	$\neq 0$		
0	$\neq 0$	0		
0	$\neq 0$	$\neq 0$		
$\neq 0$	0	0		
$\neq 0$	0	$\neq 0$		
$\neq 0$	$\neq 0$	0		
$\neq 0$	$\neq 0$	$\neq 0$		

Answer: Writing out the definition of x, we apply $\mathbf{M}$ n times and plug in the eigenvalues:

$$\begin{aligned} \mathbf{x} &= \alpha\mathbf{v}_1 + \beta\mathbf{v}_2 + \gamma\mathbf{v}_3 \\ \mathbf{M}^n\mathbf{x} &= \mathbf{M}^n(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2 + \gamma\mathbf{v}_3) \\ &= \alpha\mathbf{M}^n\mathbf{v}_1 + \beta\mathbf{M}^n\mathbf{v}_2 + \gamma\mathbf{M}^n\mathbf{v}_3 \\ &= \lambda_1^n\alpha\mathbf{v}_1 + \lambda_2^n\beta\mathbf{v}_2 + \lambda_3^n\gamma\mathbf{v}_3 \\ &= 1^n\alpha\mathbf{v}_1 + 2^n\beta\mathbf{v}_2 + \left(\frac{1}{2}\right)^n\gamma\mathbf{v}_3 \end{aligned}$$

Taking the limit as $n \to \infty$:

$$\begin{aligned} \lim_{n\to\infty} \mathbf{M}^n(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2 + \gamma\mathbf{v}_3) &= \lim_{n\to\infty} \left(1^n\alpha\mathbf{v}_1 + 2^n\beta\mathbf{v}_2 + (\tfrac{1}{2})^n\gamma\mathbf{v}_3\right) \\ &= \lim_{n\to\infty} (\alpha\mathbf{v}_1 + 2^n\beta\mathbf{v}_2) \end{aligned}$$

Notice how after applying the matrix $\mathbf{M}$ a large number times, the contribution from $\mathbf{v}_3$ diminishes to nothing because $(\frac{1}{2})^\infty \to 0$.

If $\beta \neq 0$, the system will explode because $2^\infty \to \infty$

If $\alpha \neq 0$, the system has a nonzero steady state component, and so if the system converges, it will converge to $\alpha\mathbf{v}_1$.

α	β	γ	Converges?	$\lim_{n\to\infty} \mathbf{M}^n\mathbf{x}$
0	0	$\neq 0$	Yes	$\mathbf{0}$
0	$\neq 0$	0	No	-
0	$\neq 0$	$\neq 0$	No	-
$\neq 0$	0	0	Yes	$\alpha\mathbf{v}_1$
$\neq 0$	0	$\neq 0$	Yes	$\alpha\mathbf{v}_1$
$\neq 0$	$\neq 0$	0	No	-
$\neq 0$	$\neq 0$	$\neq 0$	No	-

Meta/Answer: (5min)

2. Dive into the Stage

David Copperfield has invented a new illusion! In his new trick, he teleports from one place on a three-dimensional stage to another. The teleporter is represented by the 3×3 matrix $\mathbf{T}$: if David starts at 3D location $\mathbf{s}[t]$, he ends up at

$$\mathbf{T}\mathbf{s}[t] = \mathbf{s}[t+1].$$

David says the matrix $\mathbf{T}$ is his secret, but he can tell us its eigenvalue-eigenvector pairs:

$$\lambda_1 = 1, \quad \mathbf{v}_1 = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix} \qquad \lambda_2 = \frac{1}{3}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix} \qquad \lambda_3 = 13, \quad \mathbf{v}_3 = \begin{bmatrix} -3 \\ 5 \\ 0 \end{bmatrix}$$

(a) David starts at $\mathbf{s}[0] = \begin{bmatrix} 8 \\ 2 \\ -5 \end{bmatrix}$. Write $\mathbf{s}[0]$ as a linear combination of the eigenvectors of $\mathbf{T}$.

Answer:

$$\begin{bmatrix} 8 \\ 2 \\ -5 \end{bmatrix} = a \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix} + b \begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix} + c \begin{bmatrix} -3 \\ 5 \\ 0 \end{bmatrix}$$

Writing this out as a system:

$$\begin{aligned} -a + 3b - 3c &= 8 \\ 2a + 5c &= 2 \\ 4a - 3b &= -5 \end{aligned}$$

Adding the first and third equations eliminates b: $3a - 3c = 3$, so $a = 1 + c$. Substituting into the second equation gives $2(1+c) + 5c = 2$, so $c = 0$ and $a = 1$. Finally, the third equation gives $3b = 4a + 5 = 9$, so $b = 3$.
Therefore $a = 1,\ b = 3,\ c = 0$, i.e. $\mathbf{s}[0] = \mathbf{v}_1 + 3\mathbf{v}_2$.

For the rest of the problem, assume David starts at the same place as in part (a): $\mathbf{s}[0] = \begin{bmatrix} 8 \\ 2 \\ -5 \end{bmatrix}$.

(b) Where will he be at $t = 50$?

Answer: Using the decomposition from the previous part and applying $\mathbf{T}$ repeatedly:

$$\begin{aligned} \mathbf{s}[0] &= \mathbf{v}_1 + 3\mathbf{v}_2 \\ \mathbf{s}[1] &= \mathbf{T}\mathbf{s}[0] = \mathbf{T}\mathbf{v}_1 + 3\mathbf{T}\mathbf{v}_2 = \lambda_1 \mathbf{v}_1 + 3\lambda_2 \mathbf{v}_2 \\ \mathbf{s}[2] &= \mathbf{T}\mathbf{s}[1] = \lambda_1^2 \mathbf{v}_1 + 3\lambda_2^2 \mathbf{v}_2 \\ &\vdots \\ \mathbf{s}[t] &= \lambda_1^t \mathbf{v}_1 + 3\lambda_2^t \mathbf{v}_2 = \mathbf{v}_1 + 3\left(\frac{1}{3}\right)^t \mathbf{v}_2 \end{aligned}$$

So at $t = 50$,

$$\mathbf{s}[50] = \mathbf{v}_1 + 3^{-49}\mathbf{v}_2 = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix} + 3^{-49} \begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix} \approx \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}.$$

Since $3^{-49} \approx 4 \times 10^{-24}$, the $\mathbf{v}_2$ component has essentially vanished by $t = 50$. Even though $\lambda_3 = 13$, it never shows up because $\mathbf{s}[0]$ has no $\mathbf{v}_3$ component ($c = 0$).

(c) David's final trick is that his show will last forever. Will he converge to a single location after infinite time steps? Justify your answer.

Answer: Yes. From the previous part, $\mathbf{s}[t] = \mathbf{v}_1 + 3\left(\frac{1}{3}\right)^t \mathbf{v}_2$. In the limit as $t \to \infty$, the first term stays at $\mathbf{v}_1$ and the second term goes to $\mathbf{0}$, so

$$\lim_{t\to\infty} \mathbf{s}[t] = \mathbf{v}_1 = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}.$$

Observe that if $\mathbf{s}[0]$ had any nonzero $\mathbf{v}_3$ component ($c \neq 0$), the 13^t term would blow up and he would not converge.

(d) Would convergence still occur if the initial position had a small nonzero component along $\mathbf{v}_3$? Explain.

Answer: No. If the initial position is $\mathbf{s}[0] = \mathbf{v}_1 + 3\mathbf{v}_2 + \varepsilon\mathbf{v}_3$ with $\varepsilon \neq 0$, then

$$\mathbf{s}[t] = \mathbf{v}_1 + 3\left(\frac{1}{3}\right)^t \mathbf{v}_2 + \varepsilon 13^t \mathbf{v}_3.$$

The magnitude of the last term grows without bound, however small ε is. Thus the perturbed trajectory does not converge.

UCB EE 66, Fall 2026, Discussion 7B, All Rights Reserved. This may not be publicly shared without explicit permission.